

Module - 4 - ODE of higher Order.

Linear D.E of second and higher order with constant coefficients

- A Differential Equation in which the dependent variable and its derivatives appears only in first degree is called a linear D.E.

A.D.E of the form

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = \phi(x)$$

is called a linear D.E of n^{th} order with constant coefficients, where $a_1, a_2, \dots, a_n$ are all constants.

- If $\phi(x) = 0$, then the eq. so obtained is called a linear homogeneous D.E and the solution of such an equation is called the complementary function CF of eqⁿ ①.

Using the notation of diff. operators.

$$D = \frac{d}{dx}, \quad D^2 = \frac{d^2}{dx^2}, \quad \dots, \quad D^n = \frac{d^n}{dx^n} \quad \text{eq ① becomes}$$

$$\text{If } \phi(x) = 0, \quad (D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n) y = 0$$

□ Method of finding complementary functions:

- i) Find the Auxiliary Eq. (A.E) - $f(m) = 0$
- ii) The following 3 cases arises in eqⁿ

case i) If the roots of the A.E are real and distinct, say $m_1, m_2, \dots, m_n$, then the general solution of eqⁿ (2) is

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}.$$

case ii) If the auxiliary has real and equal roots say $m_1 = m_2$, the G.S will be

$$y = (C_1 + C_2 x) e^{m_1 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}.$$

case iii) If two of the roots of the A.E $f(m) = 0$ are imaginary, then they will occur in pairs. Let $\alpha \pm i\beta$ be a pair of imaginary root, then the general solution of eqⁿ (2) is

$$y = e^{\alpha x} [C_1 \cos \beta x + C_2 \sin \beta x] + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}$$

* If these two pairs of imaginary roots, say $\alpha \pm i\beta, \delta \pm i\delta$ then the general sol will be.

$$y = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x) + e^{\delta x} (C_3 \cos \delta x + C_4 \sin \delta x) + C_5 e^{m_5 x} + \dots + C_n e^{m_n x}.$$

* If the A.E $f(m) = 0$ has two equal pairs of imaginary roots say $\alpha \pm i\beta$ and $\alpha \pm i\beta$ occurring twice, then the general solution of eqⁿ (2) is

$$y = e^{\alpha x} [(C_1 + C_2 x) \cos \beta x + (C_3 + C_4 x) \sin \beta x] + C_5 e^{m_5 x} + \dots + C_n e^{m_n x}.$$

Q Solve the following differential equations.

1) $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = 0$

Solⁿ:

$D^3y - 3D^2y + 3Dy - y = 0$ where $D = \frac{d}{dx}$

$(D^3 - 3D^2 + 3D - 1)y = 0$ is of the form.

$f(D)y = 0$

where $f(D) = D^3 - 3D^2 + 3D - 1$.

A.E. $f(m) = 0$

$m^3 - 3m^2 + 3m - 1 = 0$

By inspection $m=1$ is a root.

$m=1$	1	-3	3	-1
	0	1	-2	1
	1	-2	1	0

$\therefore m^2 - 2m + 1 = 0$

$m = 1, 1$

3 roots are 1, 1, 1

$\therefore$ the general solution of the given D.E is

CF = $y = (C_1 + C_2x + C_3x^2)e^{1x}$

2) $(4D^4 - 4D^3 - 23D^2 + 12D + 36)y = 0$

Solⁿ:

$$A.E = 4m^4 - 4m^3 - 23m^2 + 12m + 36 = 0$$

By inspection $m = 2$ is a root

$m = 2$	4	-4	-23	12	36
	0	8	8	-30	-36
	4	4	-15	-18	0

$$\therefore 4m^3 + 4m^2 - 15m - 18 = 0 \quad \text{--- (1)}$$

By inspection $m = 2$ is a root

$m = 2$	4	4	-15	-18
	0	8	24	18
	4	12	9	0

$$\therefore 4m^2 + 12m + 9 = 0 \quad \text{--- (2)}$$

$$4m^2 + 6m + 6m + 9 = 0$$

$$2m(2m+3) + 3(2m+3) = 0$$

$$(2m+3)(2m+3) = 0$$

$$(2m+3) = 0$$

$$m = \frac{-3}{2}$$

$$(2m+3) = 0$$

$$m = \frac{-3}{2}$$

4 roots are $2, 2, \frac{-3}{2}, \frac{-3}{2}$

General solⁿ is given by

$$y = CF = (c_1 + c_2x)e^{2x} + (c_3 + c_4x)e^{-\frac{3}{2}x}$$

3). $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$

Soln:

$$AE = (4m^4 - 8m^3 - 7m^2 + 11m + 6) = 0$$

By inspection $m = -1$ is a root

$m = -1$	4	-8	-7	11	6
	0	-4	12	-5	-6
	4	-12	5	6	0

$$4m^3 - 12m^2 + 5m + 6 = 0$$

By inspection $m = 2$ is a root

$m = 2$	4	-12	5	6
	0	8	-8	-6
	4	-4	-3	0

$$4m^2 - 4m - 3 = 0$$

$$4m^2 - 6m + 2m - 3 = 0$$

$$2m(2m - 3) + 1(2m - 3) = 0$$

$$(2m + 1)(2m - 3) = 0$$

$$m = -\frac{1}{2}$$

$$m = \frac{3}{2}$$

The roots are $-1, 2, -\frac{1}{2}, \frac{3}{2}$.

General Solⁿ of the given D.E is

$$CF = C_1 e^{-x} + C_2 e^{2x} + C_3 e^{-1/2 x} + C_4 e^{3/2 x}$$

4) $(D^4 - m^4)y = 0$

Solⁿ:

A.E. $= p^4 - m^4 = 0$

[$\because$ not replaced by m since question involves m]

$\Rightarrow (p^2)^2 - (m^2)^2 = 0$

$(p^2 + m^2)(p^2 - m^2) = 0$

$\therefore p^2 + m^2 = 0$

$p^2 = -m^2$

$p = \pm \sqrt{-m^2}$

$p = \pm mi$

$p^2 - m^2 = 0$

$p^2 = m^2$

$p = \pm \sqrt{m^2}$

$p = \pm m$

Roots are $+m, -m, \pm mi, -mi$

General Solⁿ of the Given D.E. is

$y = CF = C_1 e^{mx} + C_2 e^{-mx} + e^{0x} [C_3 \cos mx + C_4 \sin mx]$

5) $(D^4 + 64)y = 0$

Solⁿ A.E. $= m^4 + 64 = 0$

$(m^2)^2 + 8^2 = 0$

w.k.T $a^2 + b^2 = (a+b)^2 - 2ab$

Hence $a = m^2, b = 8$

$(m^2 + 8)^2 - 2 \times m^2 \times 8 = 0$

$(m^2 + 8)^2 - 16m^2 = 0$

$(m^2 + 8) - (4m)^2 = 0$

$(m^2 + 8 + 4m)(m^2 + 8 - 4m) = 0$

$$m^2 + 4m + 8 = 0$$

$$m = \frac{-4 \pm \sqrt{16 - 4 \times 1 \times 8}}{2 \times 1}$$

$$= \frac{-4 \pm \sqrt{-16}}{2}$$

$$= \frac{-4 \pm 4i}{2}$$

$$m = -2 \pm 2i$$

$$m^2 - 4m + 8 = 0$$

$$m = \frac{+4 \pm \sqrt{16 - 4 \times 1 \times 8}}{2 \times 1}$$

$$= \frac{+4 \pm \sqrt{-16}}{2}$$

$$= \frac{+4 \pm 4i}{2}$$

$$m = 2 \pm 2i$$

Discriminant Formula
 $d = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

4 roots are: $-2 \pm 2i$, $2 \pm 2i$

4) Solⁿ: General is

$$y = e^{-2x} (C_1 \cos 2x + C_2 \sin 2x) + e^{2x} (C_3 \cos 2x + C_4 \sin 2x)$$

6). $\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 0$ given that $x(0) = 0$, $\frac{dx}{dt}(0) = 15$

Solⁿ: Given $\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 6x = 0$

$$(D^2 + 5D + 6)x = 0, \text{ where } D = \frac{d}{dt}$$

AE: $m^2 + 5m + 6 = 0$

$$m^2 - 3m - 2m + 6 = 0$$

$$m = -2, -3$$

General Solⁿ is $= C_1 e^{-2t} + C_2 e^{-3t}$ — (1)

differentiate w.r.t. t

$$\frac{dx}{dt} = C_1 x^{-2} + C_2 x^{-3}$$

i.e. $\frac{dx}{dt} = -2C_1 e^{-2t} - 3C_2 e^{-3t}$ — (2)

Condⁿ ① $x=0$ at $t=0$

eqⁿ ① $0 = C_1 e^0 + C_2 e^0$
 $0 = C_1 + C_2$ ——— ③

Condⁿ ② $\frac{dx}{dt} = 15$ at $t=0$

eqⁿ ② $15 = -2C_1 e^0 - 3C_2 e^0$
 $15 = -2C_1 - 3C_2$ ——— ④

Solving ③ and ④,

$$\begin{array}{rcl} C_1 + C_2 & = & 0 \quad \rightarrow \times 2. \\ -2C_1 - 3C_2 & = & 15 \end{array}$$

$$\begin{array}{rcl} 2C_1 + 2C_2 & = & 0 \\ \text{Add (+)} \quad -2C_1 - 3C_2 & = & 15 \\ \hline -C_2 & = & 15 \end{array}$$

$$\boxed{C_2 = -15}$$

$$\boxed{C_1 = 15}$$

Substituting C_1 and C_2 in eqⁿ ①

$$\boxed{x = 15e^{-2t} - 15e^{-3t}}$$

□ Inverse. Differential operators and the Particular Integral Solⁿ of Non-Homogeneous linear D.E with constant coefficients.

□ Consider a non-homogeneous linear D.E

$$\frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_n y = \phi(x)$$

i.e. $(D^n + a_1 D^{n-1} + \dots + a_n) y = \phi(x)$

$$\Rightarrow f(D) y = \phi(x)$$

→ Form the A.E $f(m) = 0$, find the roots and write the CF.

→ find P.I by P.I = $\frac{1}{f(D)} \phi(x)$, where $\frac{1}{f(D)}$ is the inverse.

Differential operators.

* Write the general solⁿ as $y = CF + P.I$

□ Particular Solⁿ of $f(D)y = \phi(x)$ (or) $P.I = y_p = \frac{\phi(x)}{f(D)}$

Type of P.I	Method of getting the P.I
1) $\frac{e^{ax}}{f(D)}$ (Also applicable for any constant $K = K \cdot e^{ax}$, e^{ax+b} , $a^x = e^{\log a(x)}$) $\sin ax$, $\cos ax$ by using their definitions.	$\frac{e^{ax}}{f(a)}$, $f(a) \neq 0$ (D is replaced by a) if $f(a) = 0$, then $x \cdot \frac{e^{ax}}{f'(a)}$; $f'(a) \neq 0$ if $f'(a) = 0$, then $x^2 \cdot \frac{e^{ax}}{f''(a)}$; $f''(a) \neq 0$ $f(x)$
2) $\frac{\sin ax}{f(D^2)}$, $\frac{\cos ax}{f(D^2)}$ Also applicable for $\sin(ax+b)$, $\cos(ax+b)$ (D^2 is replaced by $-a^2$) If $f(-a^2) = 0$, $\sin ax$, $\cos bx$, $\sin ax$, $\sin bx$, $\cos ax$ $\cos bx$ by changing into sum	$\frac{\sin ax}{f(-a^2)}$, $\frac{\cos ax}{f(-a^2)}$, where $f(-a^2) \neq 0$ then $x \cdot \frac{\sin ax}{f'(-a^2)}$, $x \cdot \frac{\cos ax}{f'(-a^2)}$
3) $\frac{\phi(x)}{f(D)}$, where $\phi(x)$ is a polynomial in x .	Divide $\phi(x)$ by $f(D)$ by writing $\phi(x)$ in desc. part of x and $f(D)$ is asc. power of D . Quotient = P.I as remainder will be zero.

Solve the following D.E.

1) $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 6e^{3x}$

Sol. $(D^2 - 6D + 9)y = 6e^{3x}$ where $D = \frac{d}{dx} : f(D) = D^2 - 6D + 9$

A.E = $f(m) = 0$ $m^2 - 6m + 9 = 0$ $(m-3)^2 = 0$
 $m-3 = 0$; $m-3 = 0$
 $m(3, 3)$

$\therefore$ CF = $y(C_1 + C_2x)e^{3x}$

To find P.I. = $\frac{p(x)}{f(D)}$ P.I. = $\frac{6e^{3x}}{D^2 - 6D + 9}$

$D \rightarrow a=3$ deno = 3
P.I. = $\frac{6 \cdot x \cdot e^{3x}}{2D-6}$

$D \rightarrow a=3$ deno = 0
P.I. = $\frac{3 \cdot 6x^2 \cdot e^{3x}}{2!} \rightarrow$ P.I. = $3x^2 e^{3x}$

20 Solution is $y = CF + PI = (C_1 + C_2x)e^{3x} + 3x^2 e^{3x}$

2) $(D^2 - 4D + 13)y = e^{3x} \cosh 2x + 3^x$

A.E = $f(m) = 0$ here $f(D) = D^2 - 4D + 13$
 $m^2 - 4m + 13 = 0$

$m = \frac{4 \pm \sqrt{16 - 4 \times 1 \times 13}}{2 \times 1} = \frac{4 \pm \sqrt{-36}}{2}$
 $m = \frac{4 \pm 6i}{2}$ $m = 2 \pm 3i$

30 CF = $e^{2x} [C_1 \cos 3x + C_2 \sin 3x]$

$$P.I = \frac{Q(x)}{f(D)} = \frac{e^{3x} \cosh 2x + 3^x}{D^2 - 4D + 13} = P_1 + P_2 \rightarrow (1)$$

$$P_1 = \frac{1}{D^2 - 4D + 13} \cdot e^{3x} \cosh 2x$$

$$= \frac{1}{D^2 - 4D + 13} \cdot e^{3x} \left\{ \frac{e^{2x} + e^{-2x}}{2} \right\}$$

$$= \frac{1}{2} \cdot \frac{1}{D^2 - 4D + 13} \left\{ e^{5x} + e^x \right\} =$$

$$= \frac{1}{2} \left[\frac{1}{D^2 - 4D + 13} \cdot e^{5x} + \frac{1}{D^2 - 4D + 13} \cdot e^x \right]$$

$$= \frac{1}{2} \left[\frac{e^{5x}}{5^2 - 20 + 13} + \frac{e^x}{1 - 4 + 13} \right] = \frac{1}{2} \left[\frac{e^{5x}}{18} + \frac{e^x}{10} \right]$$

$$P_2 = \frac{1}{D^2 - 4D + 13} \cdot 3^x$$

$$= \frac{1}{D^2 - 4D + 13} \cdot e^{\log 3^x} = \frac{1}{D^2 - 4D + 13} \cdot e^{x \log 3}$$

$$D \rightarrow a = \log 3$$

$$P_2 = \frac{1}{(\log 3)^2 - 4(\log 3) + 13} \cdot e^{x \log 3} = \frac{3^x}{(\log 3)^2 - 4 \log 3 + 13}$$

$$\therefore P.I = P_1 + P_2 = \frac{1}{2} \left[\frac{e^{5x}}{18} + \frac{e^x}{10} \right] + \frac{3^x}{(\log 3)^2 - 4 \log 3 + 13}$$

General Solution.

$$y = e^{2x} [C_1 \cos 3x + C_2 \sin 3x] + \frac{1}{2} \left[\frac{e^{5x}}{18} + \frac{e^x}{10} \right] + \frac{3^x}{(\log 3)^2 - 4 \log 3 + 13}$$

$$3) \frac{d^3 y}{dx^3} + 6 \frac{d^2 y}{dx^2} + 11 \frac{dy}{dx} + 6y = e^x + 1.$$

$$\text{sol}^n: (D^3 + 6D^2 + 11D + 6)y = e^x + 1$$

$$A.E = m^3 + 6m^2 + 11m + 6 = 0$$

$$\boxed{m = -1, -2, -3}$$

$$CF = C_1 e^{-x} + C_2 e^{-2x} + C_3 e^{-3x}$$

$$PI = \frac{1}{D^3 + 6D^2 + 11D + 6} \{e^x + 1\}$$

$$= \frac{e^x}{D^3 + 6D^2 + 11D + 6} + \frac{1 \cdot e^{0x}}{D^3 + 6D^2 + 11D + 6}$$

$$D \rightarrow a = 1$$

$$D \rightarrow a = 0$$

$$PI = \frac{e^x}{1+6+11+6} + \frac{1 \cdot e^{0x}}{0+0+0+6} = PI = \frac{e^x}{24} + \frac{1}{6}$$

$$\text{General solution } y = CF + PI$$

$$\boxed{y = C_1 e^{-x} + C_2 e^{-2x} + C_3 e^{-3x} + \frac{e^x}{24} + \frac{1}{6}}$$

$$4) \frac{d^3 y}{dx^3} - \frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} - 4y = \sinh(2x+3)$$

$$\text{sol}^n: (D^3 - D^2 + 4D - 4)y = \sinh(2x+3)$$

$$A.E = m^3 - m^2 + 4m - 4 = 0$$

$$\rightarrow m = +1 \pm 2i$$

$$\boxed{CF = C_1 e^x + e^{0x} [C_2 \cos 2x + C_3 \sin 2x]}$$

$$PI = \frac{1}{D^3 - D^2 + 4D - 4} \sinh(2x+3)$$

$$= \frac{1}{D^3 - D^2 + 4D - 4}$$

$$\left\{ \frac{e^{2x+3} - e^{-(2x+3)}}{2} \right\}$$

$$PI = \frac{1}{2} \left[\frac{e^{2x+3}}{D^3 - D^2 + 4D - 4} - \frac{e^{-2x+3}}{D^3 - D^2 + 4D - 4} \right]$$

$$D \rightarrow a = 2$$

$$D \rightarrow a = -2$$

$$PI = \frac{1}{2} \left[\frac{e^{2x+3}}{8-4+8-4} - \frac{e^{-(2x+3)}}{-8-4-8-4} \right]$$

$$= \frac{1}{2} \left[\frac{e^{2x+3}}{8} + \frac{e^{-(2x+3)}}{24} \right]$$

$\therefore$ General solⁿ $y = CF + PI$

$$y = C_1 e^x + C_2 \cos 2x + C_3 \sin 2x + \frac{1}{2} \left[\frac{e^{2x+3}}{8} + \frac{e^{-(2x+3)}}{24} \right]$$

□ Problems on Type 2: Sinax/Cosax

Solve the following O.E

1) $y'' - 4y' + 4y = 8 \cos 2x$
 solⁿ. $(D^2 - 4D + 4)y = 8 \cos 2x$

$$A.E = m^2 - 4m + 4 = 0 \rightarrow (m-2)^2 = 0$$

$$m = 2, 2$$

$$CF = (C_1 + C_2 x) e^{2x}$$

$$PI = \frac{f(x)}{f(D)} = \frac{8 \cos 2x}{D^2 - 4D + 4}$$

$$D^2 \rightarrow -a^2 = -2^2 = -4$$

$$PI = 8 \cdot \frac{1}{-4-4D+4} \cos 2x$$

$$= 8 \cdot \frac{1}{-4D} \cos 2x$$

$$= -2 \cdot \frac{1}{D} \cos 2x$$

$$\left[\begin{array}{l} D = \frac{d}{dx} = \text{diff. operator} \\ \frac{1}{D} = \text{inverse diff operator} \\ \quad \text{integration} \end{array} \right]$$

$$= -2 \int \cos 2x \, dx.$$

$$= -2 \times \frac{\sin 2x}{2} = -\sin 2x.$$

$$\text{Solution } y = CF + PI = (C_1 + C_2 x) e^{2x} - \sin 2x.$$

2) $(D^2 - 3D + 2)y = \sin 2x.$
 A.E $= m^2 - 3m + 2 = 0$
 $m = 1, 2$

$$CF = C_1 e^x + C_2 e^{2x}$$

$$PI = \frac{\phi(x)}{f(D)} = \frac{1}{D^2 - 3D + 2} (\sin 2x)$$

$$\therefore D^2 \rightarrow -a^2 = -2^2 = -4$$

$$PI = \frac{1}{-4 - 3D + 2} (\sin 2x) = \frac{1}{-3D - 2} (\sin 2x) = \frac{1}{-(3D + 2)} (\sin 2x)$$

$$= -\frac{1}{(3D + 2)(3D - 2)} \{ \sin 2x \} = \frac{-3D - 2}{(3D)^2 - 4} \{ \sin 2x \}$$

$$= \frac{-3D - 2}{9D^2 - 4} \{ \sin 2x \} \quad D^2 \rightarrow -a^2 = -2^2 = -4$$

$$= \frac{-3D - 2}{(9 \times -4) - 4} \{ \sin 2x \} = \frac{1}{4D} (3D - 2) (\sin 2x)$$

$$PI = \frac{1}{4D} \{ 3D (\sin 2x) - 2 \sin 2x \} = \frac{1}{4D} \{ 3 \times 2 \cos 2x - 2 \sin 2x \}$$

$$= \frac{1}{4D} \{ 6 \cos 2x - 2 \sin 2x \} = \frac{1}{4D} \{ 3 \cos 2x - \sin 2x \}$$

$$= \frac{1}{2D} \{ 3 \cos 2x - \sin 2x \}$$

General solution $\cdot y = CF + PI.$

$$y = C_1 e^x + C_2 e^{2x} + \frac{1}{2D} \{ 3 \cos 2x - \sin 2x \}$$

$$3) (D^2 - 4D + 4)y = e^{2x} + \sin x.$$

Sol:- AE: $m^2 - 4m + 4 = 0$
 $m = 2, 2$

$$CF = (C_1 + C_2 x) e^{2x}.$$

$$PI = \frac{\phi(x)}{f(D)} = \frac{1}{D^2 - 4D + 4} \{e^{2x} + \sin x\}$$

$$PI = P_1 + P_2 \quad \text{--- (1)}$$

Consider, $P_1 = \frac{e^{2x}}{D^2 - 4D + 4}$

$$D \rightarrow a = 2 \quad \text{denominator} = 0$$

$$P_1 = x \cdot \frac{1}{2D - 4} e^{2x}.$$

$$D \rightarrow a = 2 \quad \text{denominator} = 0$$

$$P_1 = x^2 \cdot \frac{1}{2} e^{2x}.$$

$$P_1 = \frac{x^2 e^{2x}}{2}$$

Consider $P_2 = \frac{1}{D^2 - 4D + 4} \{\sin x\}$

$$D^2 \rightarrow -a^2 = -1^2 = -1$$

$$P_2 = \frac{1}{-1-4D+4} \cdot \sin x$$

$$= \frac{1}{3-4D} \{ \sin x \}$$

$$P_2 = \frac{1}{(3-4D)} \cdot \frac{(3+4D)}{(3+4D)} \{ \sin x \}$$

$$= \frac{(3+4D)(\sin x)}{3^2 - (4D)^2}$$

$$= \frac{1}{9-16D^2} \times (3+4D)(\sin x)$$

$$D^2 \rightarrow -a^2 = -1^2 = -1$$

$$P_2 = \frac{1}{9-(16x-1)} (3+4D)(\sin x)$$

$$= \frac{1}{25} [3\sin x + 4D(\sin x)]$$

$$\boxed{D = \frac{d}{dx}}$$

$$PI = \frac{x^2 e^{2x}}{2} + \frac{1}{25} (3\sin x + 4\cos x)$$

General Solⁿ is $y = CF + PI$.

$$\boxed{y_1 = (c_1 + c_2 x) e^{2x} + \frac{x^2 e^{2x}}{2} + \frac{1}{25} (3\sin x + 4\cos x)}$$

$$4) (D^3 - 1)y = 3\cos 2x + e^{-2x}$$

$$\text{Sol}^n: \therefore AE = m^3 - 1$$

$$m = 1 \left| \begin{array}{cccc} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 0 \end{array} \right.$$

$$m^2 + m + 1 = 0$$

$$m = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$= \frac{-1 \pm \sqrt{-3}}{2}$$

$$m = \frac{-1 \pm i\sqrt{3}}{2}$$

$$CF = C_1 e^x + e^{-\frac{1}{2}x} \left[C_2 \cos \frac{\sqrt{3}}{2} x + C_3 \sin \frac{\sqrt{3}}{2} x \right]$$

$$PI = \frac{1}{f(D)} \cdot \phi(x)$$

$$= \frac{1}{D^3 - 1} \cdot \{3\cos 2x + e^{-2x}\}$$

$$\Rightarrow P_1 + P_2 \rightarrow \textcircled{1}$$

$$\text{Consider } P_1 = \frac{1}{D^2 - 1} \cdot 3\cos 2x$$

$$D^2 \rightarrow -a^2 = -2^2 = -4$$

$$P_1 = \frac{1}{D \cdot D^2 - 1}$$

$$P_1 = \frac{3 \cdot 1}{-4D - 1} \cdot \cos 2x$$

$$= \frac{-3 \cdot 1}{4D + 1} \cdot \cos 2x$$

$$= \frac{-3 \cdot 1}{(4D + 1)(4D - 1)} \cos 2x$$

$$= \frac{-3 \cdot (4D - 1)}{16D^2 - 1} \cos 2x$$

$$D^2 \rightarrow -a^2 \rightarrow = -2^2 = -4$$

$$= \frac{-3 \cdot (4D - 1)}{(16x - 4) - 1} \cos 2x$$

$$P_1 = \frac{-3}{-65} (4D - 1) \cos 2x$$

$$P_1 = \frac{3}{65} [4D(\cos 2x) - \cos 2x]$$

$$= \frac{3}{65} [4x - 2\sin 2x - \cos 2x]$$

$$= \frac{3}{65} [-8\sin 2x - \cos 2x]$$

$$P_1 = \frac{-3}{65} [8\sin 2x + \cos 2x]$$

$$P_2 = \frac{1}{D^3 - 1} \cdot e^{-2x}$$

$$D \rightarrow a = -2$$

$$P_2 = \frac{1}{(-2)^3 - 1} \cdot e^{-2x}$$

$$\Rightarrow \boxed{P_2 = \frac{-e^{-2x}}{9}}$$

General Solution

$$y = CF + PI$$

$$y = e_1 e^x + e^{-1/2x} \left[C_2 \cos \frac{\sqrt{3}}{2} + C_3 \sin \frac{\sqrt{3}}{2} \right] +$$

$$\frac{-3(8 \sin 2x - \cos 2x)}{65} - \frac{e^{-2x}}{9}$$

$$5) (D^2 + 3D + 2)y = 4 \cos^2 x$$

$$\text{Sol}^n. (D^2 + 3D + 2)y = 4 \left[\frac{1 + \cos 2x}{2} \right]$$

$$= 2(1 + \cos 2x)$$

$$= 2(e^{0x} + \cos 2x)$$

$$AE = m^2 + 3m + 2 = 0$$

$$m = -1, -2$$

$$CF = C_1 e^{-1x} + C_2 e^{-2x}$$

$$PI = \frac{\phi(x)}{f(D)}$$

$$PI = \frac{1}{D^2 + 3D + 2} \{ 2(e^{0x} + \cos 2x) \}$$

$$= P_1 + P_2 \quad \text{--- (1)}$$

$$P_1 = \frac{1}{D^2 + 3D + 2} \cdot 2e^{0x}$$

$$D \rightarrow a = 0$$

$$P_1 = \frac{1}{2} \cdot 2e^{0x}$$

$$\boxed{P_1 = 1}$$

$$P_2 = \frac{1}{D^2 + 3D + 2} \cdot 2 \cos 2x$$

$$D^2 \rightarrow -a^2 = -2^2 = -4$$

$$P_2 = \frac{1}{-4 + 3D + 2} \cdot 2 \cos 2x$$

$$= 2 \cdot \frac{1}{3D - 2} (\cos 2x)$$

$$= \frac{2}{(3D - 2)(3D + 2)} (3D + 2)$$

$$= \frac{2(3D + 2)}{9D^2 - 4} (\cos 2x)$$

$$D^2 \rightarrow -a^2 = -4$$

$$= \frac{2(3D + 2)}{(9x - 4) - 4} (\cos 2x)$$

$$P_2 = 2 \cdot \frac{1}{-40} \{ 3D(\cos 2x) + 2 \cos 2x \}$$

$$P_2 = \frac{-1}{20} [(3x - 2 \sin 2x) + 2 \cos 2x]$$

$$= \frac{-9}{20} [-3 \sin 2x + \cos 2x]$$

$$P_2 = \frac{-1}{10} [-3 \sin 2x + \cos 2x]$$

$$PI = 1 + \left(\frac{-1}{10} \right) (-3 \sin 2x + \cos 2x)$$

General Solution,

$$y = PI + CF$$

$$y = CF + PI$$

$$y = C_1 e^{-2x} + C_2 e^{-x} + 1 - \frac{1}{10} (\cos 2x + \sin 2x)$$

□ Problem on type 3 to find $PI = \frac{\phi(x)}{f(D)}$

$$1) (D^2 + 4)y = x^2 + e^{-x}$$

Solⁿ: A.E. $m^2 + 4 = 0$

$$m^2 = -4$$

$$m = \pm 2i$$

$$CF = e^{0x} [C_1 \cos 2x + C_2 \sin 2x]$$

$$PI = \frac{\phi(x)}{f(D)} = \frac{x^2 + e^{-x}}{D^2 + 4}$$

$$= \frac{x^2}{4 + D^2} + \frac{e^{-x}}{D^2 + 4} = P_1 + P_2 \quad \text{--- (1)}$$

$$2) y'' + 2y' + y = 2x + x^2$$

$$\text{Sol}^n: (D^2 + 2D + 1)y = 2x + x^2$$

$$\text{AE} \therefore m^2 + 2m + 1 = 0$$

$$(m+1)^2 = 0$$

$$m = -1, -1$$

$$\text{CF} = (C_1 + C_2 x)e^{-x}$$

$$\text{PI} = \frac{1}{f(D)} \phi(x)$$

$$\text{PI} = \frac{1}{D^2 + 2D + 1} \{2x + x^2\}$$

$$\text{PI} = \frac{x^2 + 2x}{1 + 2D + D^2}$$

$$\begin{array}{r} 1+2D+D^2 \overline{) \begin{array}{l} x^2 + 2x \\ x^2 + 4x + 2 \\ \hline -2x - 2 \\ -2x - 4 \\ \hline 2 \end{array} } \begin{array}{l} x^2 - 2x + 2 \\ \hline -2x - 2 \\ \hline 2 \end{array} \\ \hline \begin{array}{l} 2 \\ \hline 0 \end{array} \end{array}$$

$$\frac{x^2}{1} = x^2 \rightarrow Q$$

$$1 \times x^2 = x^2$$

$$2D(x^2) = 2 \times 2x = 4x$$

$$D^2(x^2) = D(4x) = 4$$

$$\frac{-2x}{1} = -2x \rightarrow Q$$

$$1 \times -2x = -2x$$

$$2D(-2x) = 2 \times -2 = -4$$

$$\text{PI} = x^2 - 2x + 2$$

General Solⁿ.

$$\Rightarrow (C_1 + C_2 x)e^{-x} + x^2 - 2x + 2$$

$$P_2 = x \cdot \frac{1}{2D} \{\cos 2x\}$$

$$= \frac{x}{2} \int \cos 2x \, dx$$

$$= \frac{x}{2} \times \frac{\sin 2x}{2}$$

$$P_2 = \frac{x \sin 2x}{4}$$

$$P_3 = \frac{1}{D^2 + 4} \{2^{-x}\}$$

$$= \frac{1}{D^2 + 4} \{e^{\log 2^{-x}}\}$$

$$= \frac{1}{D^2 + 4} e^{-x(\log 2)}$$

$$= \frac{1}{D^2 + 4} \{e^{(-\log 2)x}\}$$

$$D \rightarrow a = -\log 2$$

$$P_3 = \frac{1}{(-\log 2)^2 + 4} \cdot 2^{-x}$$

$$PI = \frac{x^2}{4} - \frac{1}{8} + \frac{x \sin 2x}{4} + \frac{1}{(-\log 2)^2 + 4} \cdot 2^{-x}$$

$$\text{General solution } y = CF + PI$$

$$y = e^{0x} \left[C_1 \cos 2x + C_2 \sin 2x \right] + \frac{x^2}{4} - \frac{1}{8} + \frac{x \sin 2x}{4} + \frac{1}{(-\log 2)^2 + 4} \cdot 2^{-x}$$

$$4) (D-2)^2 y = 8(e^{2x} + \sin 2x + x^2)$$

Soln. AE: $(m-2)^2 = 0$

$$m-2=0$$

$$m-2=0$$

$$m=2, 2$$

$$CF = (C_1 + C_2 x) e^{2x}$$

$$PI = \frac{1}{(D-2)^2} \{ 8(e^{2x} + \sin 2x + x^2) \}$$

$$= \frac{1}{D^2 - 4D + 4} \{ 8(e^{2x} + \sin 2x + x^2) \}$$

$$\Rightarrow P_1 + P_2 + P_3$$

Consider $P_1 = \frac{1}{D^2 - 4D + 4} \cdot e^{2x}$

$$D \rightarrow a=2 \quad \text{deno} = 0$$

$$P_1 = 8 \cdot x \cdot \frac{1}{2D-4} \cdot e^{2x}$$

$$D \rightarrow a=2 \quad \text{deno} = 0$$

$$P_1 = 8x^2 \cdot \frac{1}{2} e^{2x}$$

$$P_1 = 4x^2 e^{2x}$$

$$P_2 = \frac{1}{D^2 - 4D + 4} \cdot 8 \sin 2x$$

$$D^2 \rightarrow -a^2 = -2^2 = -4$$

$$P_2 = \frac{1}{-4-4D+4} \cdot 8\sin 2x$$

$$= 8 \cdot \frac{1}{-4D} \cdot \sin 2x$$

$$= -2 \int \sin 2x dx$$

$$= -2x - \frac{\cos 2x}{2}$$

$$P_2 = \cos 2x$$

$$P_3 = \frac{1}{D^2-4D+4} \{8x^2\} = \frac{8x^2}{4-4D+D^2}$$

$$\begin{array}{r} 2x^2 \\ 4-4D+D^2 \overline{) 8x^2} \\ \underline{(-) 8x^2 - 16x + 4} \\ 16x - 4 \\ \underline{(-) 16x - 16} \\ 12 \\ \underline{12} \\ 0 \end{array}$$

$$\frac{8x^2}{4} = 2x^2 \rightarrow \text{Q}$$

$$4 \times 2x^2 = 8x^2$$

$$-4D(2x^2) = -4 \times 2 \times 2x = -16x$$

$$D^2(2x^2) = D(4x) = 4$$

$$\frac{16x}{4} = 4x \rightarrow \text{Q}$$

$$-4D(4x) = -16 \times 1 = -16$$

$$\frac{12}{4} = 3 \rightarrow \text{Q}$$

$$P_3 = 2x^2 + 4x + 3$$

General solution $y = CF + PI$

$$(C_1 + C_2 x) e^{2x} + \frac{x^2}{2} 8e^{2x} + \cos 2x + 2x^2 + 4x + 3$$

□ Linear D.E with Variable coefficients

[I] I) Solution of Legendre's linear D.E

* An equation of the form

$a_0(ax+b)^2 y'' + a_1(ax+b)y' + a_2 y = \phi(x)$ is called Legendre linear eqⁿ of 2nd order.

* Take the substitution $t = \log(ax+b)$ or $e^t = ax+b$

* We have $(ax+b)y' = aDy$ where $D = d/dt$

similarly $(ax+b)^2 y'' = a^2 D(D-1)y$

$(ax+b)^3 y''' = a^3 D(D-1)(D-2)y$ etc.

* Substitute these eqn ①, which reduces to linear D.E with constant coefficients.

(II) Solution of Cauchy's linear D.E :- (Take $a=1, b=0$ in eqⁿ ①)

* An equation of the form $a_0 x^2 y'' + a_1 x y' + a_2 y = \phi(x)$ - ② is called Cauchy's linear eqⁿ of 2nd order.

* Take the substitution $t = \log x$ or $e^t = x$.

we have $xy' = Dy$ where $D = d/dt$

similarly $x^2 y'' = D(D-1)y$

$x^3 y''' = D(D-1)(D-2)y$ etc.

* Substitute these in equation ②, which reduces to linear D.E with constant coefficients.

NOTE :- Solve the linear D.E with constant coefficients to obtain $y = CF + PI$ in terms of 't'. substitute for 't' and obtain the solⁿ 'y' in terms of 'x'.

□ Problems on Cauchy's and Legendre's Linear Differential Equation

1) $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + y = \log x$

Solⁿ: Given $x^2 y'' - xy' + y = \log x$ — (1)

Eq (1) is a Cauchy's linear D.E.
Substitute $t = \log x$
 $e^t = x$

We have $xy' = Dy$ where $D = \frac{d}{dt}$
 $x^2 y'' = D(D-1)y$

Substituting in eqⁿ (1),

$$D(D-1)y - Dy + 1 \cdot y = t$$

$$(D^2 - D - D + 1)y = t$$

$$(D^2 - 2D + 1)y = t \quad \text{--- (2) where } D = \frac{d}{dt}$$

A.E $m^2 - 2m + 1 = 0$
 $m = 1, 1$

C.F. $= (c_1 + c_2 t) e^t$

P.I. $= \frac{1}{f(D)} \phi t = \frac{t}{1 - 2D + D^2}$

We'll form the division.

$$\begin{array}{r} t+2 \\ 1-2D+D^2 \overline{) y} \\ \underline{(-) x-2} \\ 2 \\ \underline{(-) 2} \\ 0 \end{array}$$

$$\begin{cases} \frac{t}{1} = t \rightarrow 0 \\ -2D(t) = -2 \times 1 = -2 \\ \frac{2}{1} = 2 \rightarrow 0 \end{cases}$$

$$PI = t+2.$$

G. solⁿ is $y = CF + PI$.

$$y = (C_1 + C_2 t) e^t + t+2$$

$$y = (C_1 + C_2 \log x) x + \log x + 2$$

$$2) x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + 9y = \sin(3 \log x) + 3x^2$$

solⁿ:- Given $x^2 y'' + xy' + 9y = \sin(3 \log x) + 3x^2$

Eq ① is Cauchy's D.E.

Substitute $t = \log x \Rightarrow e^t = x$

we have $xy' = Dy$
 $x^2 y'' = D(D-1)y$

where $D = \frac{d}{dt}$

Substitute in eq ①

$$D(D-1)y + Dy + 9y = \sin(3t) + 3(e^t)^2$$

$$(D^2 - D + D + 9)y = \sin 3t + 3e^{2t}$$

$$(D^2 + 9)y = \sin 3t + 3e^{2t} \quad \text{--- (2)} \quad \text{where } D = \frac{d}{dt}$$

$$AE = m^2 + 9 = 0$$

$$m = +3i, -3i$$

$$CF = C_1 \cos 3t + C_2 \sin 3t$$

$$PI = \frac{1}{f(D)} \phi(x) = \frac{\sin 3t + 3e^{2t}}{D^2 + 9}$$

$$\Rightarrow P_1 + P_2$$

$$P_1 = \frac{1}{D^2 + 9} \sin 3t$$

$$D^2 \rightarrow -a^2 = -3^2 = -9, \text{ deno} = 0$$

$$P_1 = t \cdot \frac{1}{2D} \cdot \sin 3t$$

$$P_1 = \frac{t}{2} \int \sin 3t dt$$

$$= \frac{t}{2} \times \frac{-\cos 3t}{3}$$

$$P_1 = \frac{-t \cos 3t}{6}$$

$$P_2 = \frac{3e^{2t}}{D^2 + 9}$$

$$D \rightarrow a = 2$$

$$P_2 = 3e^{2t}$$

$$3) \quad x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 4y = (1+x)^2$$

$$\text{Sol}^n \quad x^2 y'' - 3xy' + 4y = (1+x)^2 \quad \text{--- (1)}$$

Eqⁿ (1) is Cauchy's linear D.E

$$\text{Substitute } x = e^t \\ e^t = x$$

$$\text{We have } xy' = Dy \\ x^2 y'' = D(D-1)y$$

$$\text{where } D = \frac{d}{dt}$$

$$(1) \Rightarrow D(D-1)y - 3Dy + 4y = (1+e^t)^2$$

$$(D^2 - D - 3D + 4)y = 1 + (e^t)^2 + 2 \times 1 \times e^t$$

$$(D^2 - 4D + 4)y = 1 + e^{2t} + 2e^t \quad \text{--- (2)}$$

$$\text{A.E. } m^2 - 4m + 4 = 0$$

$$\boxed{m = 2, 2}$$

$$\text{C.F.} = (C_1 + C_2 t) e^{2t}$$

$$\text{P.I.} = \frac{1}{f(D)} \phi(t)$$

$$\text{P.I.} = \frac{1}{D^2 - 4D + 4} \{1 + e^{2t} + 2e^t\}$$

$$= P_1 + P_2 + P_3$$

$$P_1 = \frac{1}{D^2 - 4D + 4} \cdot 1 \cdot e^{0t} \rightarrow D \rightarrow a = 0$$

$$P_1 = \frac{1}{4} \cdot e^{0t}$$

$$\boxed{P_1 = \frac{1}{4}}$$

$$P_2 = \frac{1}{D^2 - 4D + 4} \cdot e^{2t}$$

$$D \rightarrow a = 2, \text{ deno} = 0$$

$$P_2 = t \cdot \frac{1}{2D - 4} \cdot e^{2t}$$

$$D \rightarrow a = 2, \text{ deno} = 0$$

$$P_2 = t^2 \cdot \frac{1}{2} e^{2t}$$

$$\boxed{P_2 = \frac{t^2 e^{2t}}{2}}$$

$$P_3 = \frac{1}{D^2 - 4D + 4} \cdot 2e^t$$

$$D \rightarrow a = 1$$

$$P_3 = \frac{1}{1 - 4 + 4} \cdot 2e^t$$

$$\boxed{P_3 = 2e^t}$$

$$PI = P_1 + P_2 + P_3$$

$$PI = \frac{1}{4} + \frac{t^2 e^{2t}}{2} + 2e^t$$

General Solution $y = CF + PI$

$$y = (C_1 + C_2 t) e^t + \frac{1}{4} + \frac{t^2 e^{2t}}{2} + 2e^t$$

$$y = (C_1 + C_2 \log x) x^{\log x} + \frac{1}{4} + \frac{t^{\log x} e^{2 \log x}}{2} + 2x$$

$$4) (1+x)^2 y'' + (1+x)y' + y = 2 \sin [\log (1+x)]$$

Sol. Eqⁿ (1) is Legendre's differential equation.

Substitute $t = \log(ax+b)$ — general

$$\text{here } a=1, b=1$$

$$t = \log(x+1)$$

$$e^t = x+1 \Rightarrow x = e^t - 1$$

we have $(x+1)y' = 1 \cdot Dy$

$$(x+1)^2 y'' = 1^2 D(D-1)y ; \quad D = \frac{d}{dt}$$

Subs. in eqⁿ (1),

$$D(D-1)y + Dy + y = 2 \sin t$$

$$(D^2 - D + D + 1)y = 2 \sin t$$

$$(D^2 + 1)y = 2 \sin t \quad \text{--- (2)}$$

$$D = \frac{d}{dt}$$

$$A.E = m^2 + 1 = 0$$

$$m = \pm i$$

$$C.F = e^{0t} [C_1 \cos t + C_2 \sin t]$$

$$P.I = \frac{1}{f(D)} \phi(t)$$

$$= \frac{1}{D^2 + 1} 2 \sin t$$

$$D^2 \longrightarrow -a^2 = -1^2 = -1, \text{ deno} = 0$$

$$P.I = t \cdot \frac{1}{2D} \cdot 2 \sin t$$

$$P.I = t \int \sin t dt$$

$$P.I = -t \cos t$$

General solution is $y = C.F + P.I.$

$$y = C_1 \cos t + C_2 \sin t - t \cos t$$

$$y = C_1 \cos [\log(x+1)] + C_2 \sin [\log(x+1)] - \log(x+1) \cos [\log(x+1)]$$

$$5) (3x+2)^2 y'' + 3(3x+2)y' - 36y = 8x^2 + 4x + 1.$$

Solⁿ: Eqⁿ ① is Legendre's D.E.

Substitute $t = \log(ax+b)$ - general
here $a=3$ $b=2$

$$\text{Sub } t = \log(3x+2)$$

$$e^t = 3x+2$$

$$e^t - 2 = 3x$$

$$x = \frac{e^t - 2}{3}$$

$$\text{we have } (3x+2)y' = 3 \cdot D y$$

$$(3x+2)y'' = 3^2 D(D-1)y$$

$$\text{Where } D = \frac{d}{dt}$$

Substitute in eq ①,

$$\Rightarrow 9D(D-1)y + 3 \times 3Dy - 36y = 8\left(\frac{e^t - 2}{3}\right) + 4\left(\frac{e^t - 2}{3}\right) + 1$$

$$\Rightarrow (9D^2 - 9D + 9D - 36)y = \frac{8}{9}(e^{2t} + 4 - 2 \times e^t \times 2) + \frac{4}{3}(e^t - 2) + 1$$

$$(9D^2 - 36)y = \frac{8}{9}e^{2t} + \frac{32}{9} - \frac{32e^t}{9} + \frac{4}{3}e^t - \frac{8}{3} + 1$$

$$(9D^2 - 36)y = \frac{8}{9}e^{2t} - \frac{20e^t}{9} + \frac{17}{9}$$

$$= 9(D^2-4)y = \frac{8}{9}e^{2t} - \frac{20}{9}e^t + \frac{17}{9}$$

$$(D^2-4)y = \frac{8}{81}e^{2t} - \frac{20}{81}e^t + \frac{17}{81} \quad \text{--- (2)}$$

$$A.E = m^2 - 4 = 0$$

$$m = \pm 2$$

$$C.F = c_1 e^{2t} + c_2 e^{-2t}$$

$$P.I = \frac{1}{f(D)} \phi(t)$$

$$P.I = \frac{1}{D^2-4} \left[\frac{8}{81}e^{2t} - \frac{20}{81}e^t + \frac{17}{81} \right]$$

$$= P_1 + P_2 + P_3$$

$$P_1 = \frac{1}{D^2-4} \left(\frac{8}{81}e^{2t} \right)$$

$$D \rightarrow a=2 \quad \text{deno} = 0$$

$$P_1 = \frac{8}{81} \cdot t \cdot \frac{1}{2 \cdot 0} \cdot e^{2t}$$

$$D \rightarrow a=2$$

$$P_1 = \frac{8}{81} \cdot t \cdot \frac{1}{2 \times 2} \cdot e^{2t}$$

$$P_1 = \frac{2}{81} t e^{2t}$$

$$P_2 = \frac{1}{D^2 - 4} \left\{ -\frac{20}{81} e^t \right\}$$

$$D \rightarrow a = 1$$

$$P_2 = \frac{1}{-3} \times \frac{-20}{81} e^t$$

$$P_2 = \frac{20}{243} e^t$$

$$P_3 = \frac{1}{D^2 - 4} \left\{ \frac{17}{81} e^{0t} \right\}$$

$$D \rightarrow a = 0$$

$$P_3 = \frac{1}{-4} \times \frac{17}{81}$$

$$P_3 = \frac{-17}{324}$$

$$PI = \frac{2}{81} te^{2t} + \frac{20}{243} e^t - \frac{17}{324}$$

$$\text{General Sol}^n. = y = CF + PI.$$

$$y = C_1 e^{2t} + C_2 e^{-2t} + \frac{2}{81} te^{2t} + \frac{20}{243} e^t - \frac{17}{324}$$

Method of Variation of Parameters

□ Working rule :

1) Given $f(D)y = \phi(x)$ where $f(D)$ is of 2nd order, write C.F. in the form $y_c = C_1 y_1 + C_2 y_2$ and assume $y = Ay_1 + By_2$ to be the complete solution of the D.E. where $A = A(x)$ & $B = B(x)$.

2) compute y_1', y_2' and Wronskian $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$

3) Assume the expression for A' and B' in the form $A' = -y_2 \frac{\phi(x)}{W}$

$B' = y_1 \frac{\phi(x)}{W}$ and similarly to obtain A & B by integration

4) sub. A, B in $y = Ay_1 + By_2$ to get the complete solution.

Solve the Linear D.E by Variation of Parameters

1) $y'' + a^2 y = \sec(ax)$

Solⁿ: $(D^2 + a^2)y = \sec(ax)$

A.E. $m^2 + a^2 = 0$

$m^2 = -a^2$

$m = \pm \sqrt{-a^2}$

$m = \pm ia$

C.F. $y_c = C_1 \cos ax + C_2 \sin ax$

$= C_1 y_1 + C_2 y_2$

$$\text{here } y_1 = \cos ax \quad ; \quad y_2 = \sin ax.$$

$$y_1' = -a \sin ax \quad ; \quad y_2' = a \cos ax$$

$$\therefore W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos ax & \sin ax \\ -a \sin ax & a \cos ax \end{vmatrix}$$

$$= a \cos^2 ax - (-a \sin^2 ax)$$

$$= a [\cos^2 ax + \sin^2 ax]$$

$$= a [1]$$

$$\boxed{W = a}$$

$$\text{Let complete sol}^n \text{ be } y = Ay_1 + By_2 \quad \text{--- (2)}$$

To find A & B

$$\text{Consider } A' = \frac{-y_2}{W} \phi(x) = \frac{-\sin ax \times \sec ax}{a}$$

$$A = \frac{-1}{a} \int \sin ax \times \sec ax \, dx + k_1$$

$$= \frac{-1}{a} \int \sin ax \cdot \frac{1}{\cos ax} \, dx + k_1$$

$$= \frac{-1}{a} \int \tan ax \, dx + k_1$$

$$= \frac{-1}{a} \int \frac{\log(\sec ax)}{a} + k_1$$

$$\boxed{A = \frac{-1}{a^2} \log(\sec ax) + k_1}$$

$$B' = \frac{y_1 \phi(x)}{w} = \frac{\cos ax \times \sec ax}{a}$$

$$= \frac{\cos ax \times 1}{a \cos ax}$$

$$B' = \frac{1}{a}$$

$$B = \int \frac{1}{a} dx + k_2$$

$$B = \frac{1}{a} \int 1 \cdot dx + k_2$$

$$B = \frac{x}{a} + k_2$$

Substituting in eq (1),

$$y = \left[\frac{-1}{a^2} \log(\sec ax) + k_1 \right] (\cos ax) + \left[\frac{x}{a} + k_2 \right] (\sin ax)$$

$$y = k_1 \cos ax + k_2 \sin ax - \frac{1}{a^2} \log(\sec ax) \times \cos ax + \frac{x}{a} \sin ax$$

2) $y'' + 4y = \tan 2x$

$$(D^2 + 4)y = \tan 2x$$

A.E. $m^2 + 4 = 0$

$$m^2 = -4$$

$$m = \pm 2i$$

$$C.F. = y_c = C_1 \cos 2x + C_2 \sin 2x$$

Here $y_1 = \cos 2x$; $y_2 = \sin 2x$
 $y_1' = -2\sin 2x$; $y_2' = 2\cos 2x$

$$\therefore W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos 2x & \sin 2x \\ -2\sin 2x & 2\cos 2x \end{vmatrix}$$

$$= 2\cos^2 2x + 2\sin^2 2x$$

$$= 2(\cos^2 2x + \sin^2 2x)$$

$$\boxed{W = 2}$$

Let complete solⁿ be $y = Ay_1 + By_2$ — (1)

To find (A) and (B)

Consider $A' = \frac{-y_2 \phi(x)}{W} = \frac{-\sin 2x \times \tan 2x}{2}$

$$A' = \frac{-\sin 2x \times \sin 2x}{2 \cos 2x}$$

$$= \frac{-\sin^2 2x}{2 \cos 2x}$$

$$= \frac{-(1 - \cos^2 2x)}{2 \cos 2x}$$

$$= \frac{-1}{2} [\sec 2x - \cos 2x]$$

$$A' = \frac{1}{2} [\cos 2x - \sec 2x]$$

$$\therefore A = \frac{1}{2} \int (\cos 2x - \sec 2x) dx + k_1$$

$$A = \frac{1}{2} \left[\frac{\sin 2x}{2} - \log \left(\frac{\sec 2x + \tan 2x}{2} \right) \right] + k_1$$

$$A = \frac{1}{4} [\sin 2x - \log (\sec 2x + \tan 2x)] + k_1$$

$$B' = \frac{y_1 \phi(x)}{w} = \frac{\cos 2x \times \tan 2x}{2}$$

$$B' = \frac{1}{2} \sin 2x$$

$$B = \frac{1}{2} \int \sin 2x dx + k_2$$

$$B = \frac{1}{2} x - \frac{\cos 2x}{2} + k_2$$

$$B = \frac{-\cos 2x}{4} + k_2$$

Sub in eqⁿ ①

$$y = \left\{ \frac{1}{4} [\sin 2x - \log (\sec 2x + \tan 2x)] + k_1 \right\} (\cos 2x) + \left[\frac{-\cos 2x}{4} + k_2 \right] (\sin 2x)$$

$$3) y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$$

$$\text{Sol: } (D^2 - 6D + 9)y = \frac{e^{3x}}{x^2}$$

$$\text{A.E. } m^2 - 6m + 9 = 0$$

$$(m-3)^2 = 0$$

$$m = 3, 3$$

$$\begin{aligned}
 CF &= (C_1 + C_2 x) e^{3x} \\
 &= C_1 e^{3x} + C_2 x e^{3x} \\
 &= C_1 y_1 + C_2 y_2 \quad (\text{say})
 \end{aligned}$$

where $y_1 = e^{3x}$
 $y_1' = 3e^{3x}$

$y_2 = x e^{3x}$
 $y_2' = x \cdot 3e^{3x} + e^{3x} \cdot 1$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{3x} & x e^{3x} \\ 3e^{3x} & x \cdot 3e^{3x} + e^{3x} \end{vmatrix}$$

$$W = e^{3x} (3x e^{3x} + e^{3x}) - (3e^{3x})(x e^{3x})$$

$$= 3x e^{6x} + e^{6x} - 3x e^{6x}$$

$$\boxed{W = e^{6x}}$$

Let the complete solⁿ be $y = A y_1 + B y_2$... is the solⁿ ①

To find A and B,

$$A' = \frac{-y_2 \phi(x)}{W} = \frac{-x e^{3x} \cdot x e^{3x}}{e^{6x} \cdot x^2}$$

$$= \frac{-x \cdot e^{6x}}{e^{6x} \cdot x^2}$$

$$\Rightarrow \frac{-1}{x}$$

$$A = -\int \frac{1}{x} dx + k_1$$

$$\boxed{A = -\log x + k_1}$$

$$B' = \frac{y_1 \phi(x)}{w} = \frac{e^{3x}}{e^{6x}} \times \frac{e^{3x}}{x^2}$$

$$B' = \frac{1}{x^2} \quad B = \int \frac{1}{x^2} dx + k_2$$

$$B = \int \frac{x^{-2}}{-} dx + k_2$$

$$B = \int \frac{x^{-2+1}}{-2+1} dx + k_2$$

$$= \frac{x^{-1}}{-1} + k_2$$

$$B = \frac{-1}{x} + k_2$$

$$y = \left[-\log x + k_1 \right] e^{3x} + \left[\frac{-1}{x} + k_2 \right] x e^{3x}$$

$$y = k_1 e^{3x} + k_2 x e^{3x} - \log x \cdot e^{3x} - e^{3x}$$

4) $y'' + y = \sec x \cdot \tan x$

Solⁿ: $(D^2 + 1)y = \sec x \cdot \tan x$

$$A.E. \cdot m^2 + 1 = 0$$

$$m = \pm i$$

$$C.F. = C_1 \cos x + C_2 \sin x$$

$$= C_1 y_1 + C_2 y_2 \text{ (say)}$$

$$y_1 = \cos x \quad ; \quad y_2 = \sin x$$

$$y_1' = -\sin x \quad y_2' = \cos x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$\begin{aligned} W &= \cos x (\cos x) - (-\sin x)(\sin x) \\ &= \cos^2 x + \sin^2 x \\ &= 1 \end{aligned}$$

$$\boxed{W = 1}$$

Let $y = Ay_1 + By_2$ be the solⁿ

To find A and B.

$$A' = \frac{-y_2 \phi(x)}{W} = \frac{-\sin x \sec x \tan x}{1}$$

$$A' = \frac{-\sin x}{\cos x} \times \frac{1}{\cos x} \times \frac{\sin x}{\cos x}$$

$$= \frac{-\sin^2 x}{\cos^2 x} = -\tan^2 x$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$\tan^2 \theta = \sec^2 \theta - 1$$

$$A' = -(\sec^2 x - 1)$$

$$A' = 1 - \sec^2 x$$

$$A = 1 - \sec^2 x$$

An Integration

$$A = \int (1 - \sec^2 x) dx + k_1$$

$$A = x - \tan x + k_1$$

$$B' = \frac{y_1 \phi(x)}{w} = \frac{\cos x \times \sec x \tan x}{1}$$

$$B' = \tan x$$

$$B = \int \tan x dx + k_2$$

$$B = \log(\sec x) + k_2$$

□

$$y = [x - \tan x + k_1] \cos x + [\log(\sec x) + k_2] \sin x$$

$$y = k_1 \cos x + k_2 \sin x + x \cos x \overset{-\sin x}{\cancel{}} + \sin x \cdot \log(\sec x)$$

$$5) y'' + 2y' + 2y = e^{-x} \sec^3 x.$$

$$\text{sol. } (D^2 + 2D + 2)y = e^{-x} \sec^3 x.$$

$$\text{A.I.E. } m^2 + 2m + 2 = 0$$

$$m = -1 \pm i$$

$$\text{CF} = e^{-x}(c_1 \cos x + c_2 \sin x)$$

$$= e^{-x} c_1 \cos x + e^{-x} c_2 \sin x$$

$$= c_1 y_1 + c_2 y_2 \quad (\text{say})$$

where $y_1 = e^{-x} \cos x$; $y_2 = e^{-x} \sin x$

$$y_1' = e^{-x} \sin x + \cos x \cdot -e^{-x}$$

$$= -e^{-x} \sin x - e^{-x} \cos x$$

$$y_2' = e^{-x} \cos x + \sin x \cdot -e^{-x}$$

$$= e^{-x} \cos x - e^{-x} \sin x$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} \cos x & e^{-x} \sin x \\ -e^{-x} \sin x - e^{-x} \cos x & e^{-x} \cos x - e^{-x} \sin x \end{vmatrix}$$

$$= e^{-x} \cos x (e^{-x} \cos x - e^{-x} \sin x) - e^{-x} \sin x (-e^{-x} \sin x - e^{-x} \cos x)$$

$$= e^{-2x} \cos^2 x - e^{-2x} \cos x \sin x + e^{-2x} \sin^2 x + e^{-2x} \sin x \cos x$$

$$= e^{-2x} (\cos^2 x + \sin^2 x)$$

$$= e^{-2x}$$

$$\boxed{W = e^{-2x}}$$

Let $y = Ay_1 + By_2$

To find A and B:

$$A' = \frac{-y_2 \phi(x)}{W} = \frac{-e^{-x} \sin x \times e^{-x} \sec 3x}{e^{-2x}}$$

$$A' = -\sin x \cdot \sec^3 x$$

$$A' = -\sin x \cdot \frac{1}{\cos x} \cdot \sec^2 x$$

$$A' = -\tan x \sec^2 x$$

$$A = - \int \tan x \sec^2 x dx + k_1$$

Subs $\tan x = t$
 $\rightarrow \sec^2 x dx = dt$

$$A = - \int t dt + k_1$$

$$= \frac{-t^2}{2} + k_1 \quad \boxed{A = \frac{-\tan^2 x}{2} + k_1}$$

$$B = \frac{y_1 \phi(x)}{w} = \frac{e^{-x} \cos x - e^{-x} \sin^3 x}{e^{-2x}}$$

$$B' = \cos x \cdot \sec^3 x$$

$$B' = \sec^2 x$$

$$B = \int \sec^2 x \cdot dx$$

$$\boxed{B = \tan x + k_2}$$

Substituting in eq (1) $y = Ay_1 + By_2$

$$= \left[\frac{-\tan^2 x}{2} + k_1 \right] e^{-x} \cos x + \left[\tan x + k_2 \right] e^{-x} \sin x$$

$$\Rightarrow \frac{-\tan^2 x}{2} \cdot e^{-x} \cos x + e^{-x} \cos x k_1 + \tan x \cdot e^{-x} \sin x + e^{-x} \sin x k_2$$

$$\Rightarrow k_1 e^{-x} \cos x + k_2 e^{-x} \sin x + \frac{\tan^2 x (e^{-x} \cos x)}{2} + e^{-x} \sin x \tan x$$

$$6) y'' + y = \frac{1}{1 + \sin x}$$

$$\text{b) } (D^2 + 1) = \frac{1}{1 + \sin x}$$

$$AE = m^2 + 1 = 0$$

$$\boxed{m^2 = -1}$$

$$CF = C_1 \cos x + C_2 \sin x.$$

$$C_1 y_1 + C_2 y_2 \text{ (say)}$$

$$y_1 = \cos x$$

$$y_1' = -\sin x$$

$$y_2 = \sin x$$

$$y_2' = \cos x$$

$$W = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$\boxed{W = 1}$$

Let the complete solⁿ be $y = Ay_1 + By_2 =$

$$A' = \frac{-y_2 \phi(x)}{W}$$

$$A' = \frac{-\sin x \times 1}{1 + \sin x}$$

$$A' = \frac{-\sin x}{1 + \sin x}$$

$$A' = \frac{-\sin x}{1 + \sin x} \times \frac{1 - \sin x}{1 - \sin x}$$

$$A' = \frac{-\sin x (1 - \sin x)}{(1 - \sin^2 x)}$$

$$A' = \frac{-\sin x + \sin^2 x}{\cos^2 x}$$

$$A' = \frac{-\sin x}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x}$$

$$A' = -\tan x \cdot \sec x + \tan^2 x$$

$$= -\sec x \tan x + (1 - \sec^2 x)$$

$$A = \int [-\sec x \tan x - 1 + \sec^2 x] dx + k_1$$

$$A = -\sec x - x + \tan x + k_1$$

$$B' = \frac{y_1 \phi(x)}{w}$$

$$= \frac{\cos x}{1} \times \frac{1}{1 + \sin x}$$

$$= \frac{\cos x}{1 + \sin x}$$

Substitutip $1 + \sin x = t$
 $\cos x dx = dt$

$$B = \int \frac{dt}{t} + k_2$$

$$B = \log t + k_2$$

$$B = \log (1 + \sin x) + k_2$$

Substituting in eq (1),

$$y = (-\sec x + \tan x - x + k_1) \cos x + (\log(1 + \sin x) + k_2) \sin x$$

$$y = -\sec x \cos x + \sin x - x \cos x + k_1 \cos x + \log(1 + \sin x) (\sin x) + k_2 (\sin x)$$

$$y = k_1 \cos x + k_2 \sin x + \sin x - x \cos x + \sin x \log(1 + \sin x)$$

SELF STUDY

□ Application to oscillation of a spring and LCR circuit.

① In a LCR circuit, the charge Q (q) on a plate of the condenser is given by $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E \sin \omega t$.

where $i = \frac{dq}{dt}$. The circuit is tuned to resonance so that

$\omega^2 = \frac{1}{LC}$. Initially $q = 0$, $i = 0$, when $t = 0$. Show that

$$q = \frac{E}{R\omega} \left[e^{-\frac{Rt}{2L}} \left(\cos pt + \frac{R}{2Lp} \sin pt \right) - \cos \omega t \right]$$

where $p^2 = \frac{1}{LC} - \frac{R^2}{4L^2}$.

Solⁿ: $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E \sin \omega t$

$$\left(LD^2 + RD + \frac{1}{C} \right) q = E \sin \omega t$$

$$A \cdot E = Lm^2 + Rm + \frac{1}{C} = 0$$

$\div L$ throughout

$$m^2 + \frac{R}{L} m + \frac{1}{LC} = 0$$

$$m = \frac{-R}{L} \pm \sqrt{\frac{R^2}{L^2} - 4 \times 1 \times \frac{1}{LC}}$$

2x1

$$m = \frac{-R}{L} \pm \sqrt{\frac{R^2}{L^2} - \frac{4}{LC}}$$

2

$$\begin{aligned} m &= \frac{-R}{2L} \pm \frac{1}{2} \sqrt{-\left(\frac{4}{LC} - \frac{R^2}{L^2}\right)} \\ &= \frac{-R}{2L} \pm \sqrt{\frac{-1}{4} \left(\frac{4}{LC} - \frac{R^2}{L^2}\right)} \\ &= \frac{-R}{2L} \pm \sqrt{-\left(\frac{1}{LC} - \frac{R^2}{4L^2}\right)} \\ &= \frac{-R}{2L} \pm \sqrt{-p^2} \end{aligned}$$

$$m = \frac{-R}{2L} \pm ip$$

$$CF = e^{-\frac{Rt}{2L}} (C_1 \cos pt + C_2 \sin pt)$$

from (1), $PI = \frac{1}{LD^2 + RD + \frac{1}{C}} \cdot E \sin \omega t$

$$D^2 \longrightarrow -a^2 = -\omega^2 = \frac{-1}{LC}$$

$$PI = \frac{1}{\left(L \times \frac{-1}{LC}\right) + RD + \frac{1}{C}} \cdot E \sin \omega t$$

$$\begin{aligned}
 P.I. &= \frac{1}{R.D} \cdot E \sin \omega t \\
 &= \frac{E}{R} \int \sin \omega t \cdot dt \\
 &= \frac{E}{R} \times \frac{-\cos \omega t}{\omega}
 \end{aligned}$$

$$\boxed{P.I. = \frac{-E}{\omega R} \cdot \cos \omega t}$$

General solⁿ is $\therefore q = C.F + P.I.$

$$q = e^{-\frac{Rt}{2L}} \{ C_1 \cos \omega t + C_2 \sin \omega t \} - \frac{E}{\omega R} \cos \omega t \quad \text{--- (2)}$$

Differentiate eq (2) w.r.t t

$$\frac{dq}{dt} = i = e^{-\frac{Rt}{2L}} \left\{ -C_1 \omega \sin \omega t + C_2 \omega \cos \omega t \right\} + \left\{ C_1 \cos \omega t + C_2 \sin \omega t \right\} \times \left(-\frac{R}{2L} \right) e^{-\frac{Rt}{2L}} + \frac{E}{\omega R} \omega \sin \omega t \quad \text{--- (2)}$$

Given cond'n 1: $q = 0$ when $t = 0$

$$(2) \Rightarrow 0 = e^0 \left[(C_1 \times 1) + (C_2 \times 0) \right] - \frac{E}{\omega R} \times 1$$

$$0 = C_1 - \frac{E}{\omega R} \quad \Rightarrow \quad \boxed{C_1 = \frac{E}{\omega R}}$$

Given Condⁿ (2): $i = 0$ when $t = 0$

$$0 = 1 \left\{ (C_1 \times 0) + (C_2 \times 1) \right\} + \left\{ (C_1 \times 1) + (C_2 \times 0) \right\} \left(-\frac{R}{2L} \times 1 \right) + 0$$

$$0 = C_2 p + C_1 \times \frac{-R}{2L}$$

$$C_2 p = \frac{C_1 R}{2L}$$

$$C_2 = \frac{C_1 R}{2pL} = \frac{R}{2pL} \times \frac{E}{\omega R}$$

$$\boxed{C_2 = \frac{E}{2pL\omega}}$$

Substituting C_1 and C_2 in (2)

$$q = e^{-\frac{Rt}{2L}} \left\{ \frac{E}{\omega R} \cos pt + \frac{E}{2pL\omega} \sin pt \right\} - \frac{E}{\omega R} \cos \omega t$$

taking $\frac{E}{\omega R}$ common

$$q = \frac{E}{\omega R} \left[e^{-\frac{Rt}{2L}} \left(\cos pt + \frac{R}{2pL} \sin pt \right) - \cos \omega t \right]$$